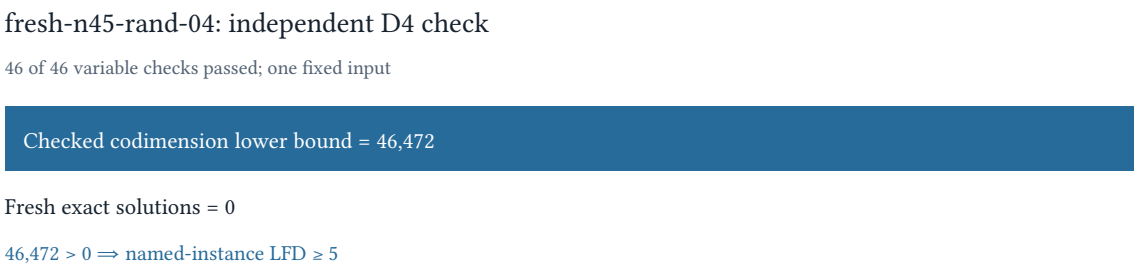
Certified n45 random04: LFD at least five

The independent D4 checker accepted all 46 variable multiplication checks for fresh-n45-rand-04. It certified codimension at least **46,472** and fresh exhaustive enumeration found **0** solutions. Thus the strict inequality establishes **LFD at least 5** for this named Boolean presentation.

The certificate is 1,045,099,360 bytes with SHA-256 e51f0a18c958179bbb15cb1b18e413f6db7a253c7119c010f6200530fa0686d5. The collector checked all 47 worker artifacts and successful terminal return; three controls passed. Public certificate and evidence readback were verified.



Admission passed after 0.007 seconds at 47,642,947,584 bytes available. Producer and checker took 2084.614 and 573.432 seconds. The original panel now has 32 terminal, 25 certified LFD-at-least-five, and no remaining inputs. Frozen source 608be655740605d96d97157006c29894332a72db; protocol 88dfb67ca54f3397a329b89ceaa4a60adc2a325ea8d24f996ffc3f7e996dc9e2.

Input n=45, k=23, seed 9620025718748421855, degree 4. Runtime command and paths are in report.md and the frozen plan. CPU time was not recorded.